

Second Semester Midterm Exam

Level: 1st Year

Material: Algorithms and Dynamic Data Structures

Date: 06/04/2026

Duration: 01h30

General guidelines

- Each exercise must be written on a separate sheet.
- Write your name, group and date on each sheet.
- Documents, smartphones, smartwatches and calculators are strictly prohibited.

Exercise 01: (05 points)

Let n be an integer number written under base 10 (see example below).

Example:

- if $n = 39$, then $(39)_{10} = (100111)_2$; $n = 23$, then $(23)_{10} = (10111)_2$.

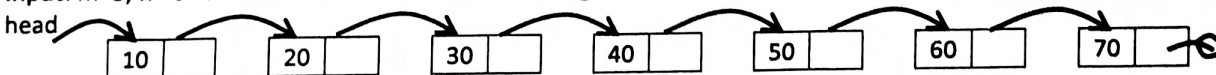
1. Propose the most suitable data structures. (0.5 point)
2. Develop the different functions and procedures that allow to convert integer n to a binary and store it into a linked linear list digit by digit. (04 points)
3. Write the main algorithm. (0.5 point)

Exercise 02: (07 points)

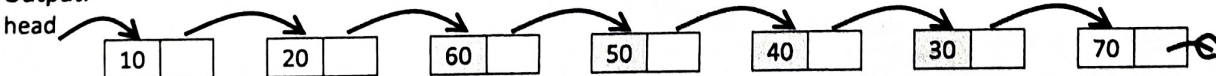
Given a linked linear list and two integer positions m and n (see example below). We need to reverse the linked linear list from position m to n .

Example:

Input: $m=3, n=6 \Rightarrow$ Linked linear list reversed starting from the node 30 to the node 60.



Output:



1. Propose the most suitable data structures. (0.5 point)
2. Write the different functions and procedures that reverse the list from position m to n . (06 points)
3. Write the main algorithm. (0.5 point)

Exercise 03: (08 points)

A Least Frequently Used (LFU) memory cache is a caching algorithm that removes the Item with the lowest access frequency when the cache reaches its maximum capacity. LFU considers both how often a page is accessed (frequency) and how recently it was used. Pages with higher access frequency are less likely to be removed. If multiple pages share the same frequency, the one that was accessed least recently (following a LRU policy) is removed.

1. Propose the most suitable data structures. (0.5 point)
2. Write the different modules that allow to manage the LFU memory cache. (07 points)
3. Write the main algorithm. (0.5 point)

Good luck